POLYP DETECTION MEDICAL REPORT

Analysis Date: June 04, 2026

VIDEO INFORMATION

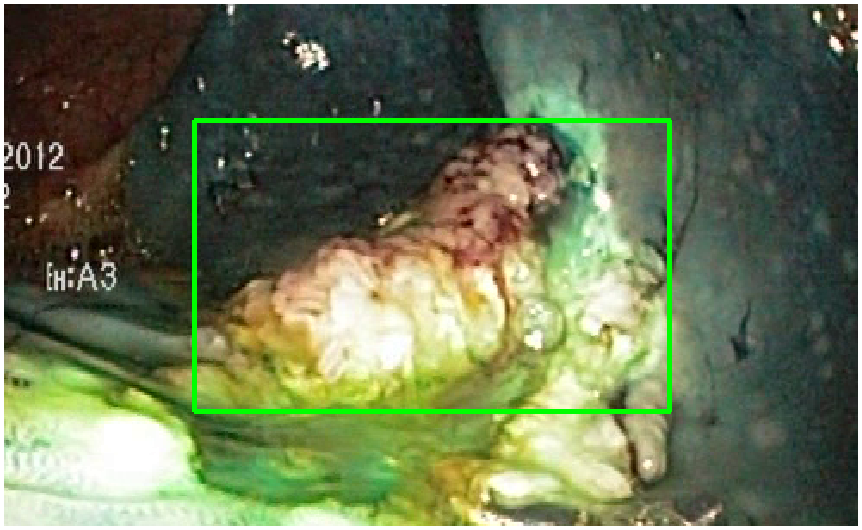
Video ID:	b7ab596e-5081-4801-b008-d97f3256caac
Total Frames:	930
Frame Rate:	25.0 fps (assumed)
Duration:	37.2 seconds

DETECTION SUMMARY

Detection Method	Count	Status
YOLO Detections	153	✓
RT-DETR Detections	615	✓
MedSAM2 Detections	550	✓
Consensus (3-Model Agreement)	7	✓
Risk Classification	**Count**	**Percentage**
HIGH RISK	0	0.0%
MEDIUM RISK	0	0.0%
LOW RISK	3	42.9%
TOTAL ANALYZED	7	100.0%

DETAILED POLYP ANALYSIS

POLYP #1 — LIFTED_POLYP [LOW_RISK] (Tracked across 89 frames)

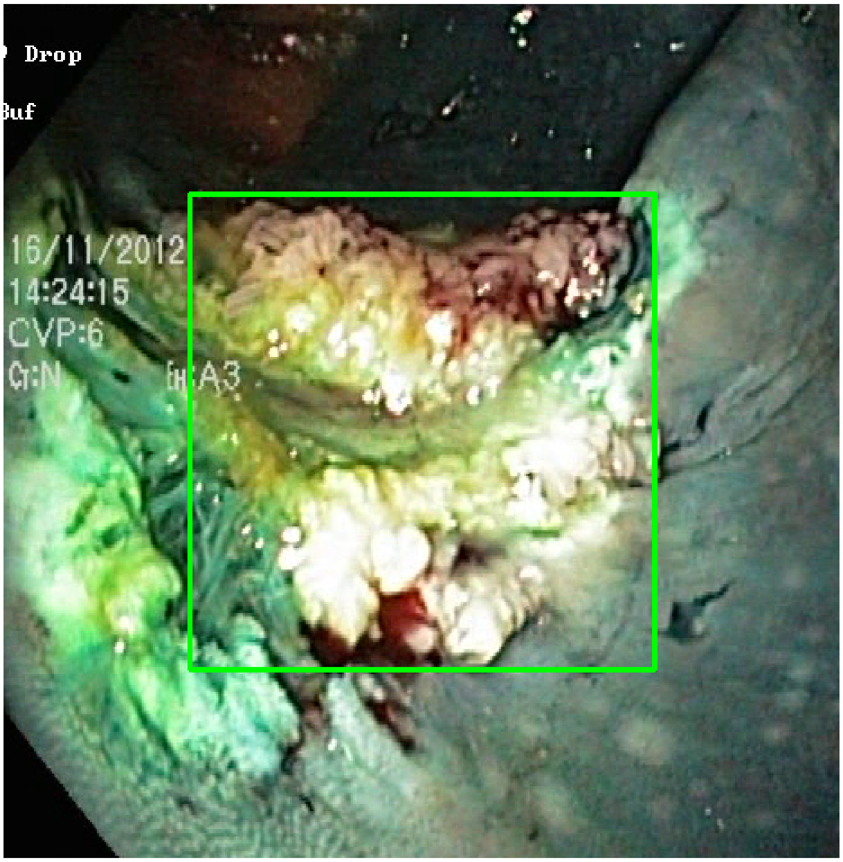


Feature	Value	Interpretation
Frame Index	5	Frame 5 of 930
Detection Method	Detector Consensus (YOLO + RT-DETR agreement) (MedSAM2 unavailable for this video)	(MedSAM2 Unavailable)
POLYP TYPE	**LIFTED_POLYP**	Type confidence: 71.1%
Redness Score	0.025	Color-based vascularization
Relative Radius	29.1%	Polyp size as % of frame diagonal
Texture Score	0.254	Surface morphology
Vessel Visibility	0.019	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	65.1%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #2 — LIFTED_POLYP [LOW_RISK] (Tracked across 246 frames)



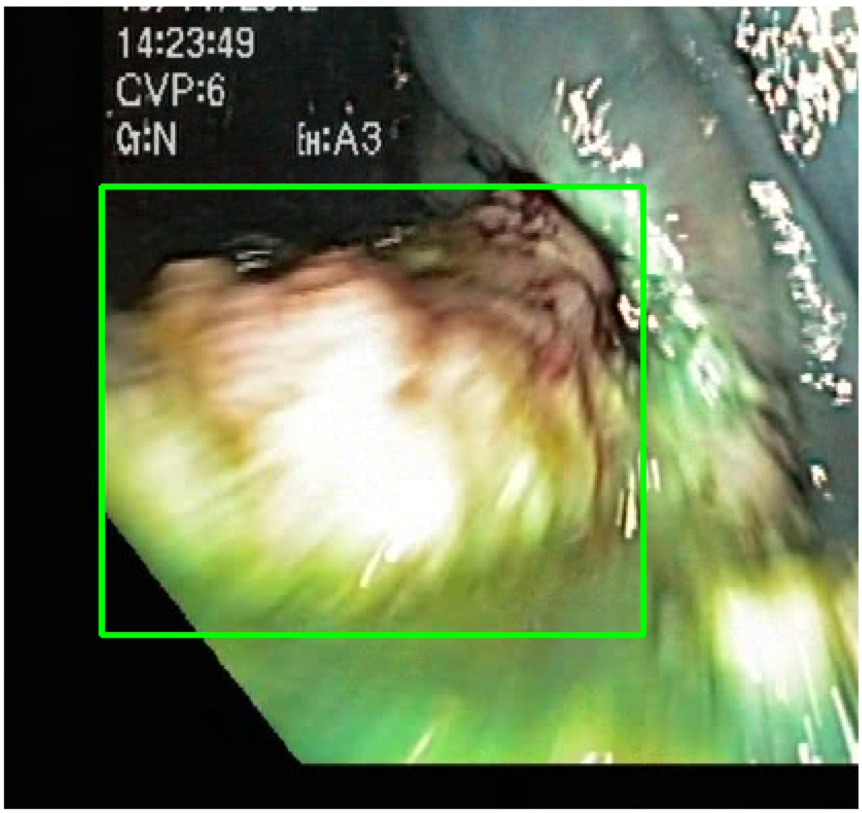
Feature	Value	Interpretation
Frame Index	827	Frame 827 of 930
Detection Method	Detector Consensus (YOLO+RT-DETR agreement) (MedSAM2 unavailable for this video)	(MedSAM2 Unavailable)
POLYP TYPE	**LIFTED_POLYP**	Type confidence: 80.3%
Redness Score	0.028	Color-based vascularization
Relative Radius	36.7%	Polyp size as % of frame diagonal
Texture Score	0.433	Surface morphology
Vessel Visibility	0.055	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	LOW_RISK	Final symbolic reasoning bucket

Expert Prediction	LOW_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	69.2%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #3 — LIFTED_POLYP [LOW_RISK] (Tracked across 20 frames)



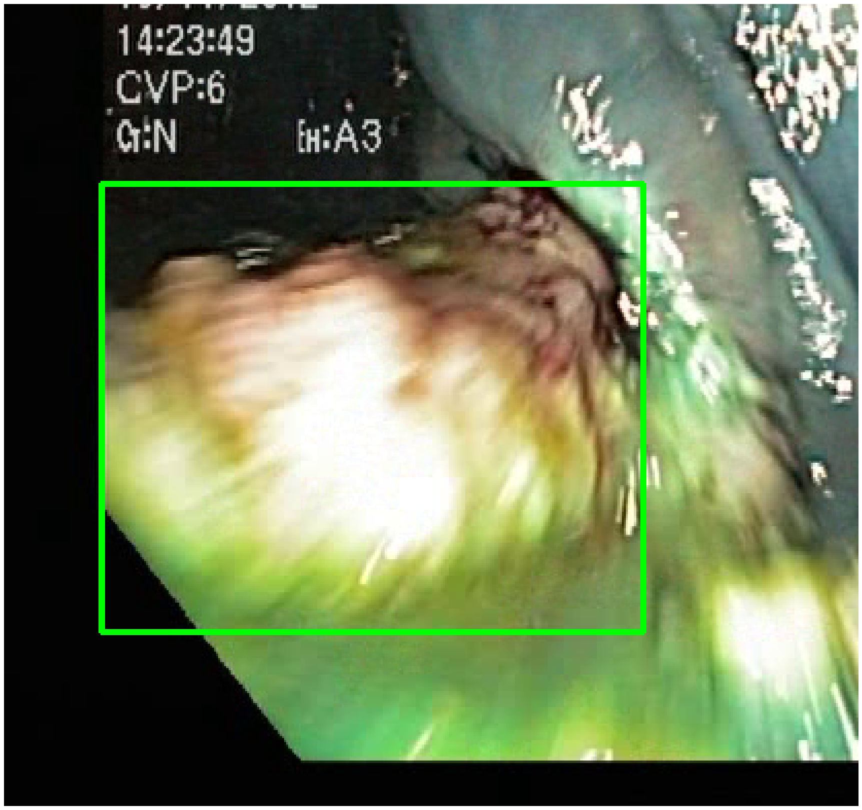
Feature	Value	Interpretation
Frame Index	178	Frame 178 of 930
Detection Method	Detector Consensus (YOLO + RT-DETR agreement (MedSAM2 unavailable for this video)	(MedSAM2 Unavailable)
POLYP TYPE	**LIFTED_POLYP**	Type confidence: 68.0%
Redness Score	0.033	Color-based vascularization
Relative Radius	30.0%	Polyp size as % of frame diagonal

Texture Score	0.160	Surface morphology
Vessel Visibility	0.015	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	66.8%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #4 — LIFTED_POLYP [LOW_RISK] (Tracked across 20 frames)



Feature	Value	Interpretation
Frame Index	178	Frame 178 of 930
Detection Method	Detector Consensus (YOLO+RT-DETR agreement (MedSAM2 unavailable for this video)	(MedSAM2 Unavailable)
POLYP TYPE	**LIFTED_POLYP**	Type confidence: 68.0%
Redness Score	0.033	Color-based vascularization
Relative Radius	30.0%	Polyp size as % of frame diagonal
Texture Score	0.160	Surface morphology
Vessel Visibility	0.015	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified

Detection Confidence	75.0%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #5 — FLAT_POLYP [MEDIUM_RISK] (Tracked across 20 frames)

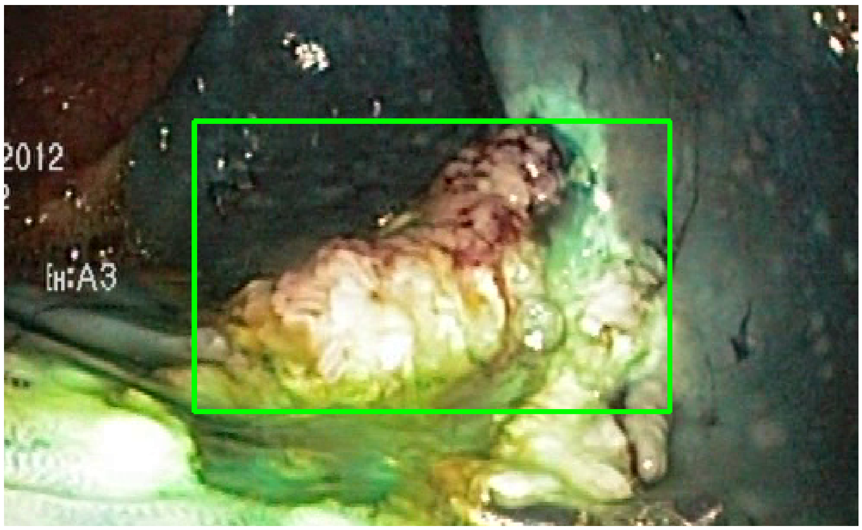
Feature	Value	Interpretation
Frame Index	854	Frame 854 of 930
Detection Method	Detector Consensus (YOLO + RT-DETR agreement) (MedSAM2 unavailable for this video)	(MedSAM2 Unavailable)
POLYP TYPE	**FLAT_POLYP**	Type confidence: 82.2%
Redness Score	0.043	Color-based vascularization
Relative Radius	26.6%	Polyp size as % of frame diagonal
Texture Score	0.398	Surface morphology
Vessel Visibility	0.139	Vascularization indicator
Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output

Clinical Class	NORMAL_MUCOSA	Normal mucosal pattern — no polyp morphology identified
Detection Confidence	74.4%	YOLO / RT-DETR / track confidence

Type Assessment: Flat mucosal lesion — chromoendoscopy or NBI assessment recommended.

Low-confidence MODERATE RISK detection - Clinical review recommended

POLYP #6 — LIFTED_POLYP [LOW_RISK] (Tracked across 50 frames)



Feature	Value	Interpretation
Frame Index	5	Frame 5 of 930
Detection Method	Detector Consensus (YOLO + RT-DETR agreement (MedSAM2 unavailable for this video)	(MedSAM2 Unavailable)
POLYP TYPE	**LIFTED_POLYP**	Type confidence: 72.8%
Redness Score	0.032	Color-based vascularization
Relative Radius	26.0%	Polyp size as % of frame diagonal
Texture Score	0.232	Surface morphology
Vessel Visibility	0.053	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output

Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	55.7%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #7 — FLAT_POLYP [MEDIUM_RISK] (Tracked across 20 frames)

Feature	Value	Interpretation
Frame Index	854	Frame 854 of 930
Detection Method	Detector Consensus (MedSAM2 unavailable)	YOLO + RT-DETR agreement (MedSAM2 unavailable for this video)
POLYP TYPE	**FLAT_POLYP**	Type confidence: 82.2%
Redness Score	0.043	Color-based vascularization
Relative Radius	26.6%	Polyp size as % of frame diagonal
Texture Score	0.398	Surface morphology
Vessel Visibility	0.139	Vascularization indicator
Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	NORMAL_MUCOSA	Normal mucosal pattern — no polyp morphology identified
Detection Confidence	64.4%	YOLO / RT-DETR / track confidence

Type Assessment: Flat mucosal lesion — chromoendoscopy or NBI assessment recommended.

Low-confidence MODERATE RISK detection - Clinical review recommended

CLINICAL IMPRESSION

Summary:
A total of 7 polyps were detected and analyzed using multi-model consensus and AI-based risk stratification.

Risk Distribution:
• HIGH RISK polyps: 0 (0.0%)
• MEDIUM RISK polyps: 0 (0.0%)
• LOW RISK polyps: 3 (42.9%)

Recommendation:
All detected polyps are classified as LOW RISK. Standard surveillance protocol is recommended.
Technical Details:
• Detection: Multi-model consensus (YOLO, RT-DETR, MedSAM2)
• Features: 444-dimensional vectors (384 SSL + 60 biomarkers)
• Classification: Cluster-specific Decision Tree experts
• Confidence: Based on deep learning + medical heuristics

Disclaimer: *This report is generated by an AI-assisted research system for academic and research purposes only. All clinical recommendations are derived from published ESGE 2017, NICE CG118/CG131, and BSG 2019 colonoscopy guidelines. This output does not constitute medical advice and must not be used as the sole basis for clinical decisions. All findings must be reviewed and confirmed by a qualified gastroenterologist or endoscopist.*

AI-Based Polyp Detection System
Generated: 2026-06-04 08:19:09